LD10 Glossary

Simple definitions for Boosting, XGBoost, CatBoost, LightGBM and Optuna

Term	Simple meaning
Boosting	A method that trains models one after another, where each new model focuses on previous errors.
Bagging	A method that trains many models independently and combines their predictions.
Random Forest	A bagging model that combines many decision trees trained with random rows and random features.
OOB validation	Out-of-bag validation. A way to estimate Random Forest performance using rows not selected by each tree.
Weak learner	A simple model that is only slightly better than guessing.
Decision stump	A very small decision tree with only one split.
AdaBoost	A boosting method that increases the weight of wrongly predicted rows.
Gradient Boosting	A boosting method where each new model learns how to reduce the loss of the current model.
Forward stagewise additive modelling	Building a model step by step by adding one new learner at a time.
Loss function	A formula that measures how wrong a model is.
Pseudo-residual	The correction direction that the next boosting model should learn.
Learning rate	Controls how strongly each new tree changes the model.
n_estimators	The number of trees or boosting rounds.
max_depth	Controls how deep each tree can grow.
Early stopping	Stops training when validation performance no longer improves.
XGBoost	A strong and regularised Gradient Boosting library for tabular data.
CatBoost	A boosting library designed to handle categorical columns well.
LightGBM	A fast boosting library that is efficient for larger tabular datasets.
GridSearchCV	A tuning method that tests all combinations from a fixed parameter grid.
Optuna	A tuning framework that searches smarter than a fixed grid by learning from previous trials.
Trial	One Optuna experiment with one set of hyperparameters.
RMSE	Root Mean Squared Error. A regression metric where lower is better.
F1-score	A classification metric balancing precision and recall.
Feature importance	A model output showing which columns were used most strongly for prediction.
Averaging ensemble	A model combination where predictions from several models are averaged.